2 **(a)** A field of force is a region of space in which a body experiences a force without physical contact with another body. [The field is produced by the presence of another body (or bodies)]. [B1]

(b) State one similarity and one difference between gravitational field and electric field.

Similarity [B1]:
- The field strength due to a point charge/mass follows an inverse square law with distance from the charge/mass.
- The potential due to a point charge/mass varies inversely with distance from the charge/mass.
- (Both G-fields and E-fields can be represented by field lines) In notes, we accept for students who memorised.

Difference [B1]:
- Gravitational fields are always attractive, whereas electric fields can be attractive or repulsive.
- Gravitational potential is always negative, whereas electric potential can be positive or negative.
- Source of gravitational field is a mass while source of electric field is a charge.

(c) **(i)** At point P, it is a gravitational neutral point where net gravitational field strength g is zero. [B1]

It is closer to the moon since the mass of the moon is smaller than the mass of the Earth. [B1]

(ii) At point P, *gravitational neutral point*.

$$g_M = g_E$$

$$G\frac{M_M}{r^2} = G\frac{M_E}{\left(3.80\times10^8 - r\right)^2} \quad \text{[M1- apply } g = G\frac{M}{r^2}\text{]}$$

$$\frac{\left(3.80\times10^8 - r\right)^2}{r^2} = \frac{5.98\times10^{24}}{7.35\times10^{22}} \quad \text{[B1]}$$

$$r = 3.79\times10^7 \text{ m} \quad \text{[A0]}$$

OR using potential method to solve,
Sum of potentials at point P (give method mark if approach is shown)

(iii) By conservation of energy

loss in KE = gain in GPE

$$\frac{1}{2}(m)(v)^2 - 0 = m(\phi_f - \phi_i)$$

$$\frac{1}{2}(v)^2 - 0 = (-1.3 - (-62.3)) \times 10^6 \quad \text{[C1]}$$

$$v = 11018$$

$$= 1.10 \times 10^4 \text{ m s}^{-1} \quad \text{[A1]}$$

The space probe needs to have sufficient energy from the Earth's surface to just reach the point P with maximum potential. Thereafter, the net gravitational force on the space probe (pointing towards Moon) will enable it to reach the Moon. [B1]

(d) **(i)** P is negative.
Near P, the electron has a positive potential energy. [M1]

Being very close to P, potential energy due to P on the electron is more significant than potential energy due to Q on the electron. (i.e. **approximately** $E_P \approx \dfrac{Q_P e}{4\pi\varepsilon_0 r}$)
For potential energy to be positive, P and electron must be of the same sign.
Hence P is negative. [A1]

OR

P is negative.
Potential energy increases towards P.[M1]

Since positive work must be done on electron to move it towards P.
Therefore, P and electron must be repelling each other.
Hence P is negative. [A1]

OR

P is negative.
Gradient of the graph is negative. [M1]

Since force points in the direction of decreasing potential energy ($F = -\dfrac{dU}{dr}$) electrostatic force on electron must be in the direction of positive r (right).
Hence P and electron must be repelling each other
Hence P is negative. [A1]

(ii) Magnitude of force = gradient of potential energy graph [C1]

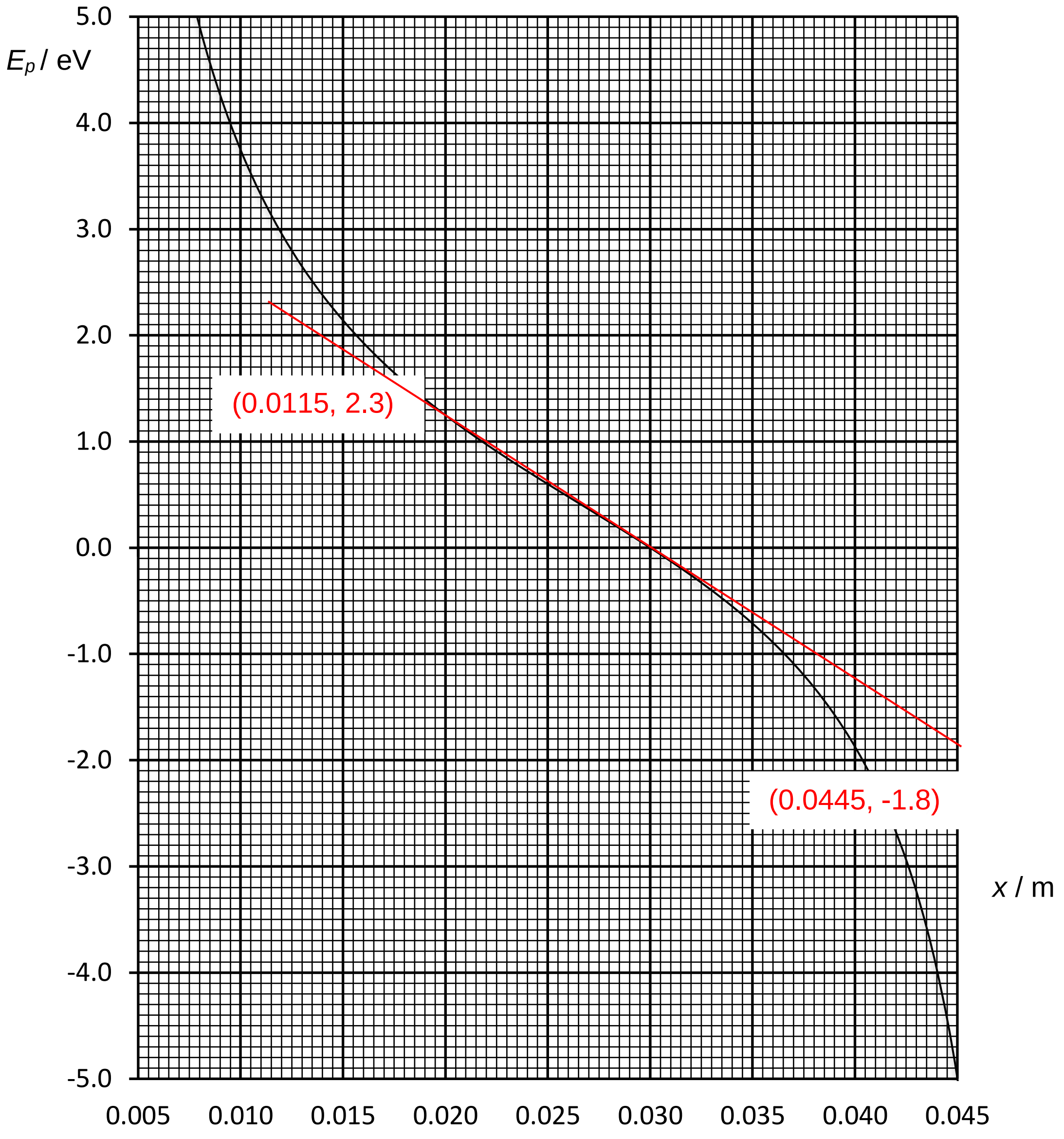
Gradient of graph (at x = 3.0 cm)

$$= \frac{2.3-(-1.8)}{(0.0115-0.0445)} = 124 \text{ eV m}^{-1}$$ [C1: gradient within range]

Accept gradient between 110 to 135 eV m^{-1}

Convert the unit to SI unit, force = 124 x 1.6 x 10^{-19} = 1.98 x 10^{-17} N
[A1:conversion to SI]

Accept a range of force due to gradient.
1.76 x 10^{-17} N
2.16 x 10^{-17} N



(iii) Curve with value of Ep near P to be lower & steeper gradient nearer to Q.